



A Crash-Resilient Contiguous File System for Embedded Systems

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Research Question

How to create a file system for flash storage that is resilient to random power loss events and stores files contiguously?

Introduction

- Lilota is an Internet of Things (IoT) framework that controls peripherals and modules by Tcl scripting
- Currently, the Lilota kernel is statically linked
- LilotaFS is a file system for SPI flash on embedded systems
- Enables dynamic linking of modules into kernel without increasing physical memory use

LilotaFS Design Constraints

- Run on flash: individual bits can change from 1 to 0, but only entire sectors (4KiB on ESP32) can change from 0 to 1
- Embedded devices can crash at any time => must be able to detect crashes and restore to a previous state
- Files are stored contiguously: single block with no breaks in between

The final goal is to **dynamically link** the Lilota kernel:

Static linking (now)

- Modules linked into kernel
- Large kernel binary
- **Includes all modules**, even if not needed

Dynamic linking

- Modules remain separate
- Smaller kernel
- Flexibility to only include modules users want

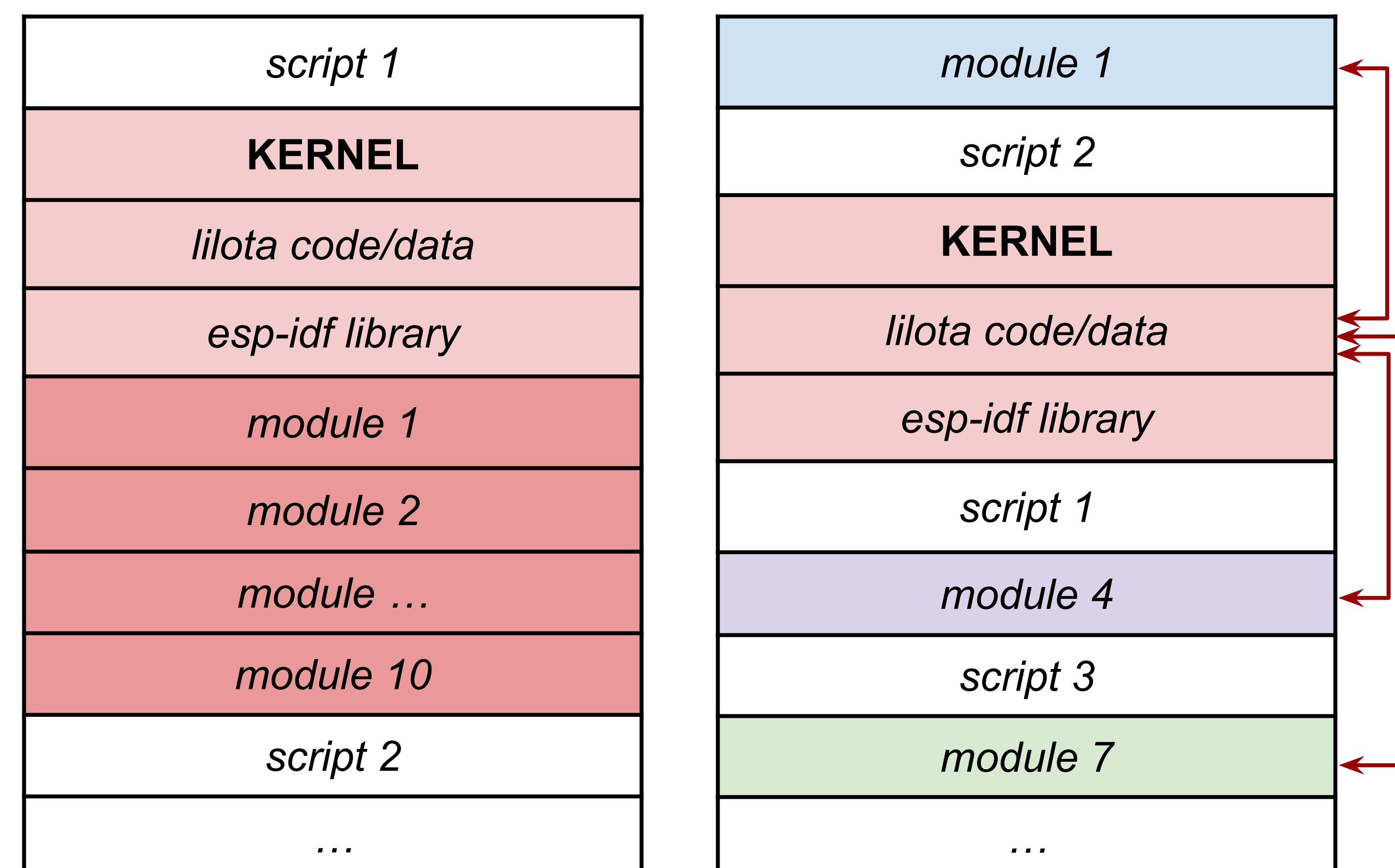


Figure 1. Comparison of current (static) vs dynamic linking.

Design

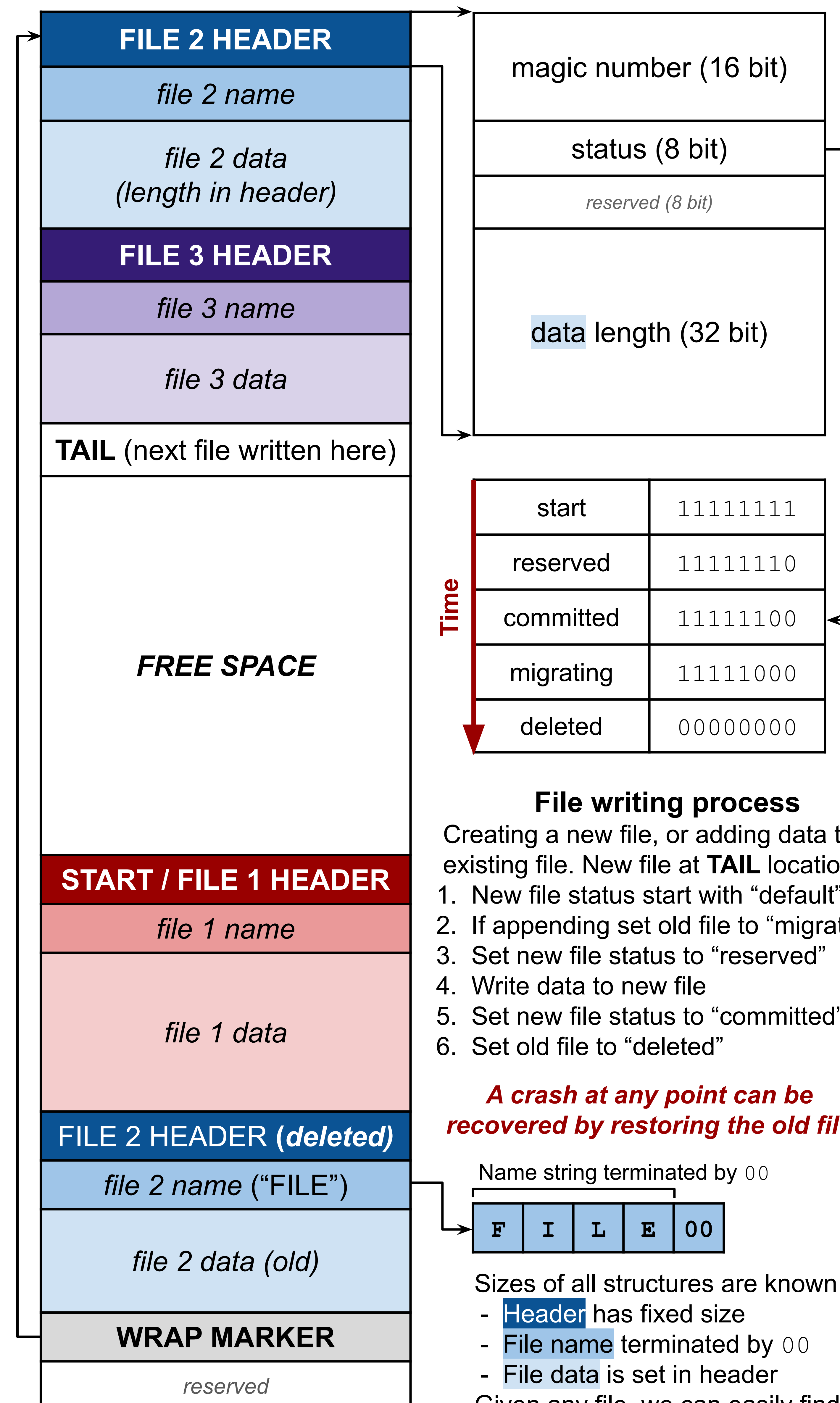
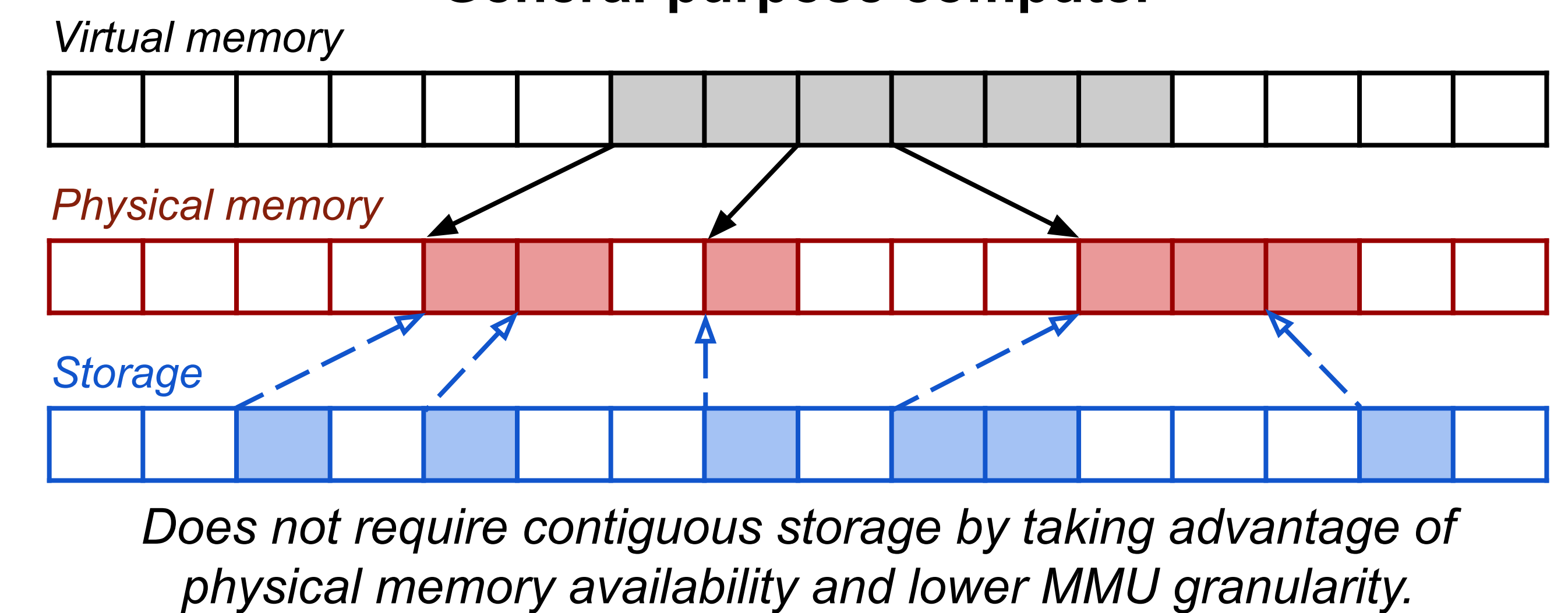


Figure 2. Sample LilotaFS image.

Motivation

- In general purpose computers: modules are copied to RAM
- Lilota has insufficient physical memory
- Skip copying to RAM by memory mapped flash directly
- Caveat: file must be contiguous to enable memory mapping

General-purpose computer



Lilota (proposed)

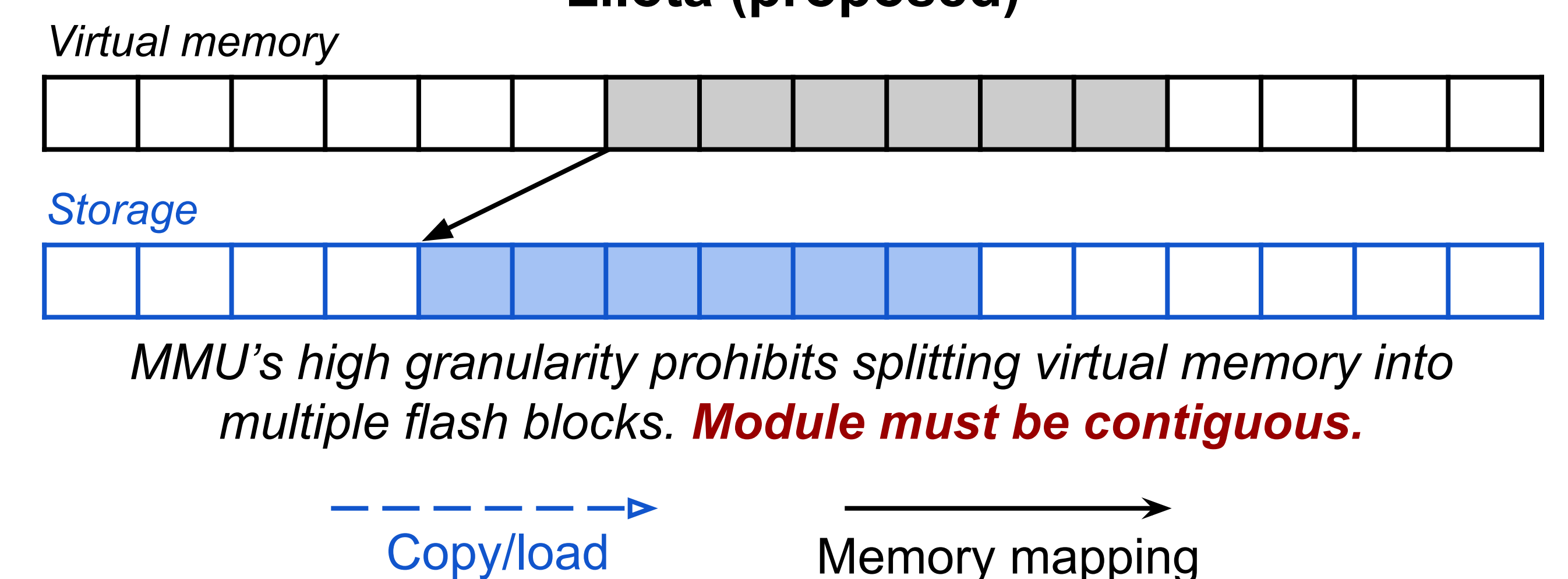


Figure 3. Comparison of memory configurations for dynamic loading.

Future Work

To reduce Lilota's binary size, separate kernel development from modules and increase resilience:

- Load Lilota modules directly from flash at runtime
- Expand LilotaFS's extreme torture test

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